

USER GUIDE

Assessment Name : Olympic Database

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Lab Group : 2 (Friday, 3pm-5pm)

Overview

This user guide provides a detailed procedure for implementing an Olympic 2024 database named Olympic_22467577 in a MySQL environment. This database is made on a Linux operating system with MySQL version 8.0.39-0ubuntu0.22.04.1. The database is designed to store data about the Olympic 2024 held in Paris, France, and serves as a valuable and valid resource for any sports research or analysis.

The guide aims to navigate users with detailed steps and clear examples so that users can create and set up the database and execute some queries and stored procedures. It is suitable for anyone who does not know about the MySQL platform to read this guide in order to use the database. It would be much better if the user had a basic knowledge of MySQL before using the database since this guide does not guide every function in MySQL. Whether you are a student or a researcher, you will gain rich data on the Olympics 2024 after using this database.

Checklist

Structure of Olympic_22467577 Database includes:

1) 8 Tables

- Athlete
➔ Shows all the information about the athletes participated
- Country
➔ Shows all the information about the countries participated
- Event
➔ Shows all the information about the events
- Venue
➔ Shows all the information about the venues used
- Medal
➔ Shows all the information about the medals
- Official
➔ Shows all the information about the officials
- Participation
➔ Shows information about which athletes joined which events
- Event_Management
➔ Shows information about which officials managed which events

2) 2 Views

- Winner
➔ Shows all the information about the athletes who won medal
- Athlete_Venue
➔ Shows all the information about which venues were used by athletes during participation

3) 3 Stored Procedures

- insAthlete
➔ Insert a new athlete into the Athlete table
- insOfficial
➔ Insert a new official into the Official table
- showCountryMedalNumber
➔ Display the countries and their total medals respectively

4) 2 Triggers

- check_age_insert_athlete
➔ Checking the inserted athlete's age validity
- change_medalcode_update_medal
➔ Checking the inserted medal's type validity

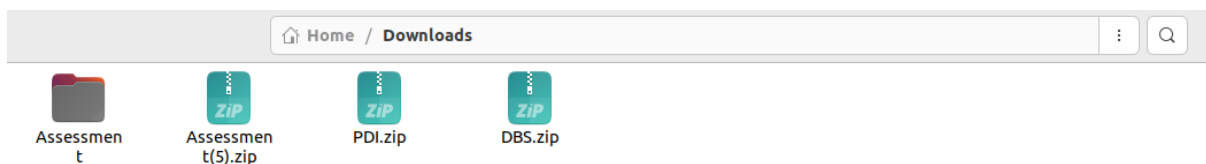
5) 1 Python Program

- ConnectOlympic.py
➔ An UI friendly program with useful function

Database User Setting-Up Guide

1. Download the YeeMingCheing_22467577_DbS2024.zip and extract the zip file.
2. Move the Assessment folder to any path.
3. Open your terminal (Press Ctrl+Alt+T).
4. Enter 'cd (Path of Assessment)/Assessment'.

Example:



➔ The extracted Assessment file is under Downloads directory

```
yee@yee-virtual-machine: ~/Downloads/Assessment
yee@yee-virtual-machine:~$ cd Downloads/Assessment
yee@yee-virtual-machine:~/Downloads/Assessment$
```

➔ Input 'cd Downloads/Assessment' to move to the Assessment Folder

5. Enter 'chmod +x setting_env.sh' to grant execute permission to setting_env.sh bash script.
6. Then, enter './setting_env.sh' to run the bash script.
7. After that, enter 'source ~/.bashrc' as prompted by the script.

```
yee@yee-virtual-machine: ~/Downloads/Assessment
yee@yee-virtual-machine:~$ cd Downloads/Assessment
yee@yee-virtual-machine:~/Downloads/Assessment$ chmod +x setting_env.sh
yee@yee-virtual-machine:~/Downloads/Assessment$ ./setting_env.sh
Please enter 'source ~/.bashrc' in the shell to update the changes in the file
yee@yee-virtual-machine:~/Downloads/Assessment$ source ~/.bashrc
yee@yee-virtual-machine:~/Downloads/Assessment$
```

Database Setting-Up Guide

1. Enter 'sudo mysql' or 'mysql -u root -p' with root password to log into the MySQL.
2. Enter 'source SettingUp.sql;' to set up the Olympic_22467577 database.

```
mysql> source SettingUp.sql;
Query OK, 0 rows affected (0.02 sec)

Query OK, 0 rows affected (0.02 sec)

Query OK, 0 rows affected (0.01 sec)

Query OK, 0 rows affected (0.01 sec)

Query OK, 10 rows affected (0.19 sec)

Query OK, 1 row affected (0.01 sec)

Database changed
Query OK, 0 rows affected, 1 warning (0.01 sec)

Query OK, 0 rows affected (0.04 sec)

Query OK, 0 rows affected, 1 warning (0.01 sec)

Query OK, 0 rows affected (0.05 sec)

Query OK, 0 rows affected, 1 warning (0.00 sec)

Query OK, 0 rows affected, 1 warning (0.04 sec)
```

Warning: semicolon (;) must be placed at the end of every MySQL command for it to execute

3. Enter 'use Olympic_22467577' to log into the database.
4. Now, you have log into the Olympic_22467577 database. Enter 'show tables;' to show all the table in the database.

```
Query OK, 0 rows affected, 1 warning (0.00 sec)

Query OK, 0 rows affected (0.01 sec)

mysql> use Olympic_22467577
Database changed
mysql> show tables;
+-----+
| Tables_in_Olympic_22467577 |
+-----+
| Athlete                     |
| Athlete_Venue              |
| Country                     |
| Event                       |
| Event_Management           |
| Medal                       |
| Official                    |
| Participation               |
| Venue                       |
| Winner                      |
+-----+
10 rows in set (0.00 sec)
```

5. You can also enter 'select * from (Table Name)' to display the data in the table. (Table Name is case sensitive)

```
mysql> select * from Country;
```

Country_Code	Name
AFG	Afghanistan
AHO	Netherlands Antilles
AIN	AIN
ALB	Albania
ALG	Algeria
AND	Andorra
ANG	Angola
ANT	Antigua and Barbuda
ARG	Argentina
ARM	Armenia
ARU	Aruba
ASA	American Samoa
AUS	Australia
AUT	Austria
AZE	Azerbaijan
BAH	Bahamas
BAN	Bangladesh

Stored Procedure Guide

This database include three stored procedure:

I. insAthlete

- Enter the following command with correct parameter
 - call insAthlete('Athlete's Name','Athlete's Gender','Athlete's Discipline','Athlete's Birth Date','Athlete's Side Job','Athlete's Country Code')
 - (If no side job, enter "")
 - e.g. call insAthlete('Yee','Male','Archery','2005-04-23','', 'IND');

```
mysql> call insAthlete('Yee','Male','Archery','2005-04-23','', 'IND');
Query OK, 1 row affected (0.01 sec)
```

1540120	FERNANDEZ Federico	Male	Handball	1989-10-17	NULL	ARG
1540143	FISCHER Guillermo	Male	Handball	1996-07-18	NULL	ARG
1540144	Yee	Male	Archery	2005-04-23	None	IND

II. insOfficial

- Enter the following command with correct parameter
 - call insOfficial('Official's Name','Official's Gender','Official's Role','Athlete's Belonged Organization')
 - e.g. call insOfficial('SecondYee','Male','Judge','HappyLand');

```
mysql> call insOfficial('SecondYee','Male','Judge','HappyLand');
Query OK, 1 row affected (0.02 sec)
```

4654138	KUSSMAUL Holger	Male	Referee	Germany
4968543	CAMPANILE Nicolas	Male	Judge	International Gymnastics Federation
4968544	SecondYee	Male	Judge	HappyLand

546 rows in set (0.00 sec)

III. showCountryMedalNumber

- Enter 'call showCountryMedalNumber();'
 - Output the countries and their total medals respectively

```
mysql> call showCountryMedalNumber();
```

Country_Code	Name	TotalMedalNumber
MEX	Mexico	3
ETH	Ethiopia	2
LCA	Saint Lucia	2
IRL	Ireland	1
JAM	Jamaica	1
PUR	Puerto Rico	1
AFG	Afghanistan	0
AHO	Netherlands Antilles	0

Trigger Guide

This database include two triggers:

I.check_age_insert_athlete

- ➔ If you insert an invalid athlete's birth date, the trigger will stop you from inserting the data

(Invalid means athlete's age is lower the minimum age which is 13)

```
mysql> call insAthlete('ThirdYee','Male','Archery','2024-04-23','','IND');
ERROR 1644 (45000): Athlete must be at least 13 years old!
mysql>
```

II.change_medalcode_update_medal

- ➔ If you update a medal type that doesn't match the medal code, the trigger will automatically change the medal type to match the medal code
- ➔ e.g. update Medal Set Medal_Code =3 where Athlete_Code=1539986;

```
mysql> update Medal Set Medal_Code =3 where Athlete_Code=1539986;
Query OK, 1 row affected (0.02 sec)
Rows matched: 1  Changed: 1  Warnings: 0
```

Then,

1537325	1	Gold	2024-08-03
1537325	2	Silver	2024-08-06
1539986	1	Gold	2024-08-03

10 rows in set (0.00 sec)



1537325	1	Gold	2024-08-03
1537325	2	Silver	2024-08-06
1539986	3	Bronze	2024-08-03

10 rows in set (0.00 sec)

Python Program Guide

1. Ensure you are in the Assessment folder. If you are still in MySQL, enter 'exit' to quit
2. Enter 'python3 ConnectOlympic.py' to run python program

```
mysql> exit
Bye
yee@yee-virtual-machine:~/Downloads/Assessment$ python3 ConnectOlympic.py

Welcome
Please select one of the following option
1)Show Athlete Table
2)Show Winner Table
3)Insert a athlete into Athlete Table
4>Delete a athlete into Athlete Table
5)Update a athlete's discipline into Athlete Table
6)QueryOne(Display information of the athlete that has a side job and was born between 1995 and 2005)
7)QueryTwo(Display information of official from a Sport Federation)
0)Exit
Enter your choice: 6
(1533136, 'BASS BITTAYE Gina Mariam', 'police officer (sub-inspector)')
(1533176, 'CAMARA Ebrahima', 'prison officer')
(1533235, 'ARELLANO Fernanda', 'student')
(1533237, 'TOSCANO Pamela', 'sport soldier')
(1533364, 'ISMAIL Mohamed', 'sport soldier')
(1533371, 'HASSAN Ibrahim', 'soldier with the Djiboutian Republican Guard')
(1533372, 'WAISS Abdi', 'sport soldier')
(1533374, 'HASSAN NOUR Samiyah', 'sport soldier')
(1535191, 'GONZALEZ Alegna', 'serves in the navy')
(1535226, 'AGUILAR CHAVEZ PEON Mariana', 'student')
(1535310, 'BLANCO Luisa', 'student')
(1535343, 'WILLARS VALDEZ Randal', 'Armed forces ')
(1535346, 'OROZCO LOZA Alejandra', 'Armed forces ')
(1535348, 'AGUNDEZ GARCIA Gabriela', 'Armed forces ')
(1535364, 'CARRILLO Duilio', 'Armed Forces ')
```

3. The Python program connects to the Olympic_22467577 database. Any changes you make in the program will affect the database
4. Follow the instructions in the program for the best user experience

Thanks for reading this user guide.
Enjoy exploring the Olympic_22467577 database!